

SOFTWARE REQUIREMENTS SPECIFICATION

Dungoo SkillsHub

AI-Powered Career Readiness Platform for Ethiopian Youth

Version 1.0

Prepared for: Hackathon Submission — MinT / AAU Startup Center, Adwa Museum

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for Dungoo SkillsHub, an AI-powered career readiness platform being developed by Dungoo Software Solutions. It is intended to guide the development team during the hackathon build and subsequent iterations, and to serve as a shared reference for judges, mentors, and future contributors evaluating the product's scope and architecture.

1.2 Scope

Dungoo SkillsHub addresses the soft-skills and employability gap facing Ethiopian youth by providing scalable, affordable, and locally relevant practice tools. The system combines peer-to-peer language exchange, an AI communication practice engine, an AI-powered video mock interview simulator, short soft-skills drills, and a persistent Skill Passport credential. The long-term roadmap extends the platform into a two-sided marketplace connecting youth with verified human experts and, eventually, employers.

This document covers the full product vision. Sections are explicitly tagged to indicate what is in scope for the initial hackathon build versus what belongs to the Phase 2 roadmap, so the reader can distinguish the demo-ready core from longer-term ambitions.

1.3 Problem Statement

Ethiopia's economy is growing at nearly 7% GDP, yet urban youth unemployment remains around 27% — a classic “jobless growth” pattern. The gap isn't ambition; it's soft skills. Rote-memorization education leaves graduates without professional English fluency, interview confidence, or workplace communication skills, even when their technical qualifications are sound. Human coaching and private training centers can close this gap, but they're cost-prohibitive and impossible to scale nationally. The result: a large pool of capable youth remains unvetted and unemployed, while employers can't find candidates who are demonstrably workplace-ready.

1.4 Solution Statement

Dungoo SkillsHub makes soft-skills, language, and interview readiness training scalable, affordable, and locally relevant for Ethiopian youth. At its core is an AI-driven practice loop: peer-to-peer language exchange (trade the language you know for the one you want, free for 40 minutes/day), an AI communication practice engine focused on professional English, an AI video mock interview simulator with real-time feedback on clarity, confidence, and STAR-method structure, and short AI-guided soft-skills drills. Every session feeds a persistent Skill Passport that tracks a user's growth over time. As the platform matures, it expands into a two-sided marketplace — verified language teachers offering paid sessions, industry professionals hosting field-matched interview rooms, and verified trainers running live group sessions on leadership and workplace skills — turning AI practice into a full ecosystem connecting youth, teachers, professionals, and eventually employers.

1.5 Unique Value Proposition

Dungoo SkillsHub is differentiated from existing tools on three fronts:

- Built for Ethiopia, not adapted from elsewhere — unlike global AI interview tools such as Yoodli and Huru, which are built for US/UK job markets and priced in USD, SkillsHub is designed ground-up for Ethiopian accents, languages, and workplace norms.
- One integrated platform, not a single point tool — unlike language-exchange apps such as Tandem and HelloTalk, SkillsHub combines peer exchange with structured AI interview and soft-skills training in a single platform.
- True multi-language matching — the matching engine reflects how Ethiopians actually communicate, across Amharic, Afaan Oromoo, and Tigrinya, not just a single language pair.

- Closes the loop on both sides of the market — youth get affordable, scalable practice; employers get a verified readiness signal through the Skill Passport.

1.6 Definitions, Acronyms, and Abbreviations

- SRS — Software Requirements Specification
- FR — Functional Requirement
- NFR — Non-Functional Requirement
- STAR — Situation, Task, Action, Result (interview answer structure)
- Skill Passport — A persistent, shareable credential aggregating a user's practiced skills and scores
- UML — Unified Modeling Language
- MVP — Minimum Viable Product

1.7 References

- Dungoo SkillsHub Problem & Solution Statement, Dungoo Software Solutions, 2026
- IEEE Std 830-1998 — Recommended Practice for Software Requirements Specifications (structural reference)
- Ethiopia urban youth unemployment and GDP growth figures, cited in project problem statement

1.8 Overview

Section 2 describes the overall product context, users, and constraints. Section 3 details functional requirements with hackathon-scope tags. Section 4 covers non-functional requirements. Section 5 presents UML diagrams (use case, class, sequence, and component/architecture). Section 6 lists supplementary requirements, and the Appendix contains scope-prioritization guidance.

2. Overall Description

2.1 Product Perspective

Dungoo SkillsHub is a new, standalone product rather than an extension of an existing system. It is designed as a mobile-first platform (Android/PWA) backed by cloud AI services, built specifically for Ethiopian languages, accents, and workplace norms — differentiating it from global tools such as Yoodli and Huru (built for US/UK markets) and language-exchange apps such as Tandem and HelloTalk (single-language-pair focus, no structured interview or soft-skills training).

2.2 Product Functions

At a high level, the system:

- Onboards youth users and captures their career goals and language profile.
- Matches peers for free, timed language exchange sessions across four languages.
- Delivers AI-guided professional English and soft-skills practice.
- Runs AI video mock interviews with real-time feedback on clarity, confidence, and STAR structure.
- Aggregates all activity into a persistent, verifiable Skill Passport.
- (Phase 2) Connects users to verified human experts and, eventually, employers, through a two-sided marketplace.

2.3 User Classes and Characteristics

User Class	Description
Youth User	Primary user; job seeker or student building employability skills. Ranges from low to moderate digital literacy; primary device is an Android phone.
Peer Learner	A youth user acting as a language-exchange partner for another user.
Content Admin	Dungoo team member managing question banks, drills, and rubrics.
Verified Expert (Phase 2)	Language teacher, industry professional, or trainer offering paid sessions.
Employer (Phase 2)	Views/verifies a candidate's Skill Passport as a hiring signal.

2.4 Operating Environment

- Client: Android mobile app and/or progressive web app, optimized for low-end devices.
- Backend: Cloud-hosted application services and AI/LLM inference (see Section 5.4 architecture diagram).
- Network: Must operate acceptably on low-bandwidth mobile connections common outside major urban centers.

2.5 Design and Implementation Constraints

- Hackathon build timeline (24–48 hours) limits the initial release to a narrow, high-impact demo path (see Appendix).
- AI feedback must support Ethiopian-accented English and, over time, Amharic, Afaan Oromoo, and Tigrinya.
- Free-tier cost structure (40 minutes/day language exchange) constrains AI inference and infrastructure cost per user.

2.6 Assumptions and Dependencies

- Assumes reasonable access to smartphones among the target youth demographic, even if bandwidth is constrained.
- Depends on third-party or in-house AI/LLM services for language scoring, speech/video analysis, and feedback generation.
- Assumes a future pipeline of verified human experts willing to join the Phase 2 marketplace.

3. Functional Requirements

Each requirement below is tagged with its hackathon build status: Core build (must work live in the demo), Lightweight/Minimal build (simplified but functional), Mockup/roadmap (shown as a static screen or slide), or Out of scope (mentioned only in the pitch).

ID	Requirement	Description	Hackathon Scope
FR-1	User Onboarding	Sign-up, profile creation, language self-assessment, career-goal/industry selection.	Minimal (build)
FR-2	Peer Language Exchange Matching	Bidirectional matching by language known/wanted across Amharic, Afaan Oromoo, Tigrinya, English; free 40-min daily sessions.	Mockup / roadmap
FR-3	AI Communication Practice Engine	Guided professional-English exercises with AI-generated corrections and prompts.	Roadmap
FR-4	AI Video Mock Interview Simulator	Role/industry question generation, video/audio capture, AI scoring on clarity, confidence, STAR structure, per-answer feedback.	Core build
FR-5	AI-Guided Soft-Skills Drills	Short structured modules with completion tracking.	Roadmap
FR-6	Skill Passport	Auto-aggregate session history, scores, and milestones into a shareable credential.	Core build (basic)
FR-7	Progress Dashboard	Visualize skill growth, completed modules, and streaks.	Lightweight build
FR-8	Marketplace Module (Phase 2)	Listings and booking for verified teachers, interview professionals, and group trainers; payments.	Pitch-deck only
FR-9	Verification System	Vet and badge human experts before marketplace listing.	Pitch-deck only
FR-10	Notifications	Reminders for sessions, streaks, and bookings.	Out of scope
FR-11	Admin/Content Management	Update interview question banks, drills, and rubrics without deployment.	Out of scope

4. Non-Functional Requirements

ID	Category	Requirement
NFR-1	Performance	AI feedback (interview/communication scoring) returned within a few seconds to preserve session flow.
NFR-2	Scalability	Must scale from pilot cohorts to nationwide usage without architectural rework.
NFR-3	Accessibility	Functions on low-bandwidth connections and low-end Android devices; lightweight video/audio handling.
NFR-4	Localization	Full multi-language support (Amharic, Afaan Oromoo, Tigrinya, English) across UI, matching, and AI feedback.
NFR-5	Data Privacy & Security	Encrypted storage/transmission of recordings, personal data, and Skill Passport records; explicit consent for recorded sessions.
NFR-6	Verifiability / Trust	Skill Passport and expert verification badges must be tamper-resistant.
NFR-7	Availability	High uptime for core AI features; peer-matching degrades gracefully when no match exists.
NFR-8	Usability	Interface accessible to users with varying digital literacy.
NFR-9	Cost Efficiency	Free core tier must remain financially sustainable at scale (AI inference, matching infra).
NFR-10	Extensibility	Architecture must cleanly support Phase 2 marketplace/payment features without a full rebuild.

5. UML Diagrams

5.1 Use Case Diagram

Illustrates the primary actors — Youth User, Peer Learner, Content Admin, and (Phase 2) Verified Expert — and their interactions with the system's core use cases. Shaded/dashed elements denote Phase 2 (marketplace) scope.

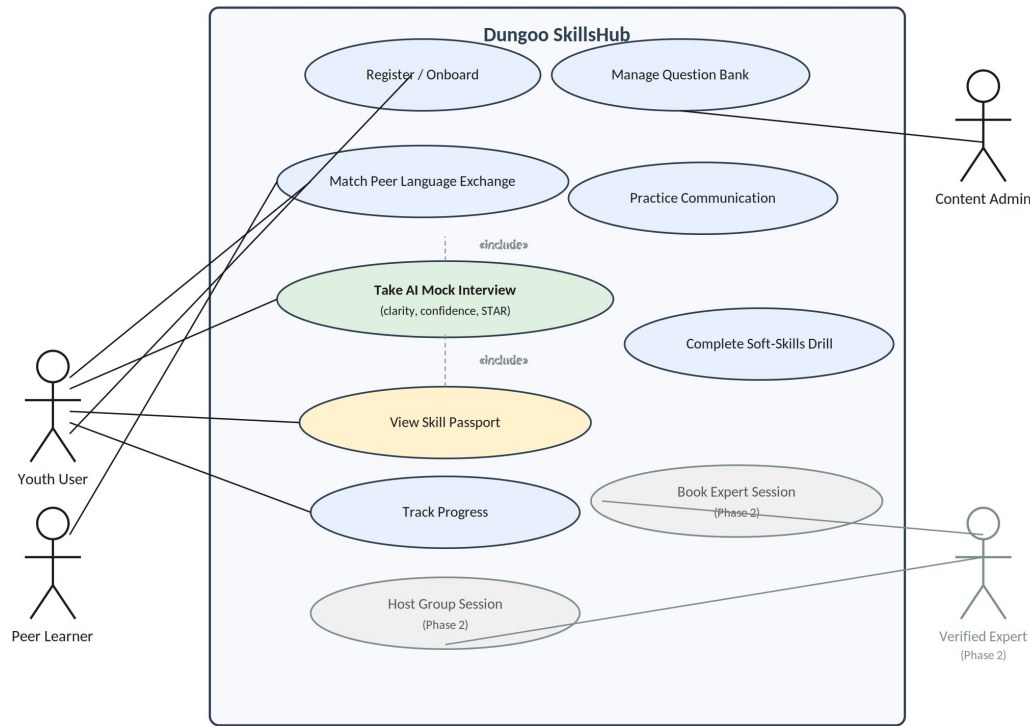
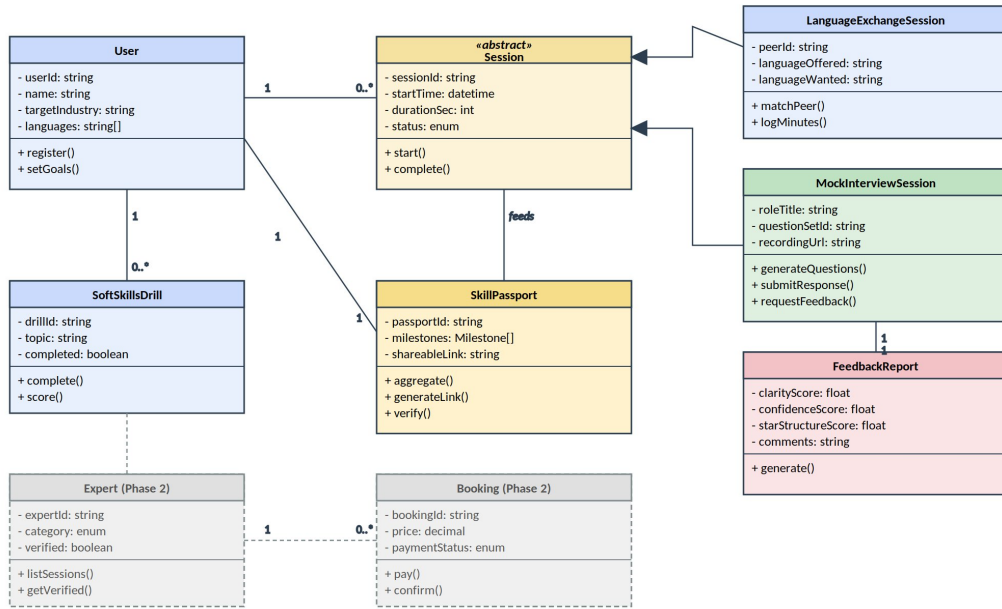


Figure 1: Use Case Diagram — Dungoo SkillsHub

5.2 Class Diagram

Models the core domain entities: User, the abstract Session type (specialized into LanguageExchangeSession and MockInterviewSession), FeedbackReport, SkillPassport, and SoftSkillsDrill. Dashed classes (Expert, Booking) represent Phase 2 marketplace entities, out of hackathon build scope.



Dashed elements = Phase 2 (marketplace) scope, out of hackathon build.

Figure 2: Class Diagram — Core Domain Model

5.3 Sequence Diagram — AI Mock Interview Flow

Traces the primary hackathon demo path end-to-end: a Youth User starts a mock interview, the Interview Service fetches role-matched questions, the AI Scoring Engine evaluates each response for clarity, confidence, and STAR structure, and the resulting scores are pushed to the Skill Passport before a final report is shown to the user.

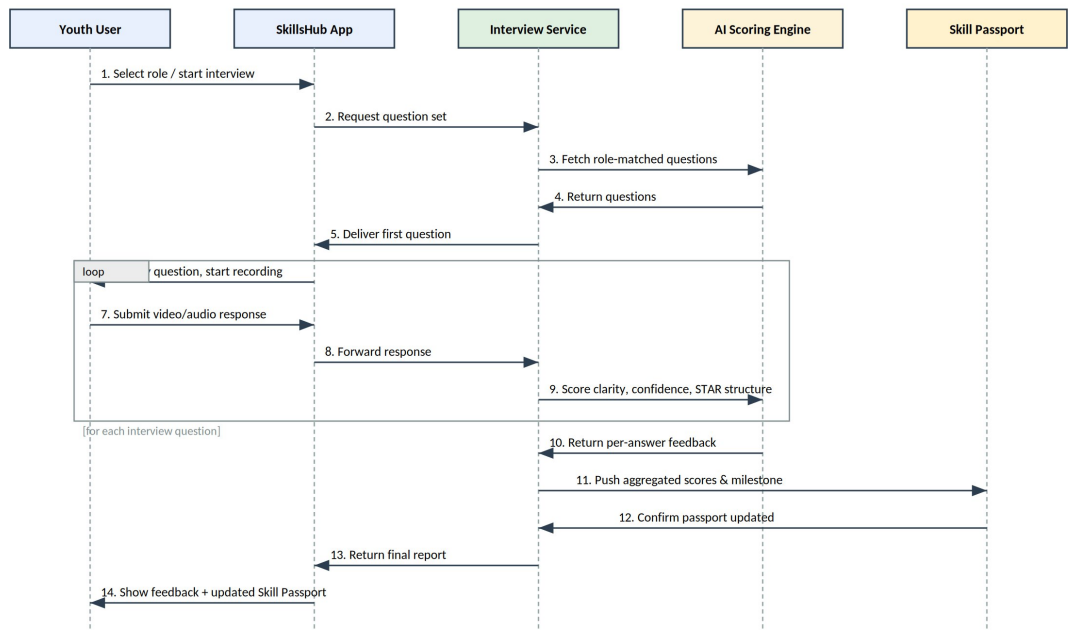


Figure 3: Sequence Diagram — AI Mock Interview Session

5.4 Component / Architecture Diagram

Shows the layered architecture: client layer (mobile app / PWA), API gateway with auth middleware, core application services, the AI Scoring Engine (with per-language models), and the encrypted data layer. The Marketplace Service is shown dashed as Phase 2 scope.

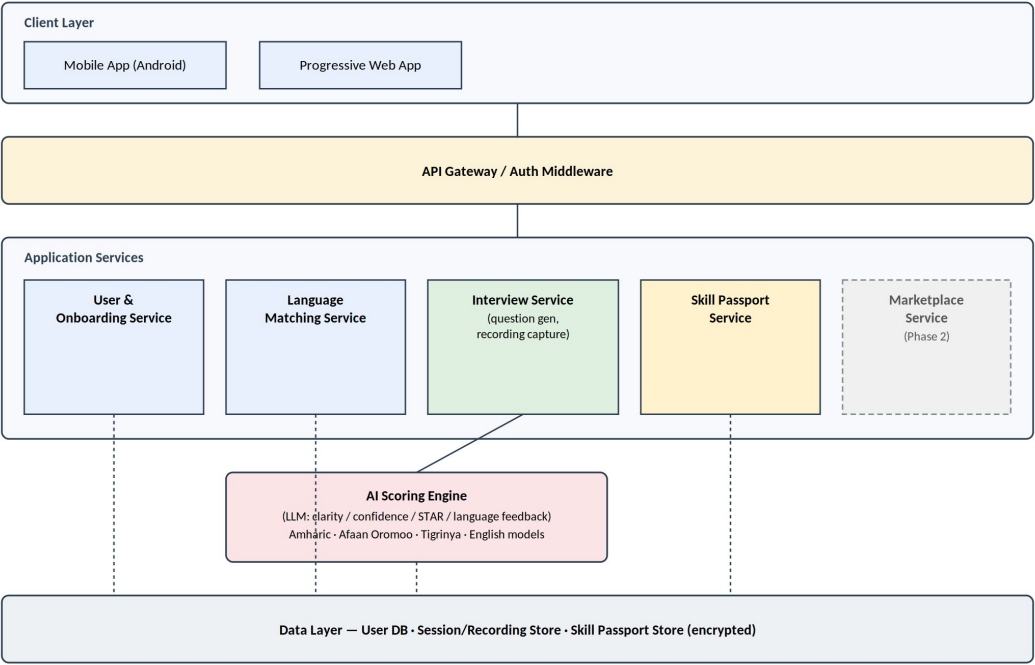


Figure 4: Component / Architecture Diagram

6. Other Requirements

6.1 Data Requirements

- User profiles, session recordings, AI-generated scores, and Skill Passport records must be stored with clear ownership per user and encrypted at rest.
- Recordings and personal data require explicit user consent before capture.

6.2 External Interface Requirements

- AI/LLM provider API(s) for language scoring, speech/video analysis, and feedback generation.
- (Phase 2) Payment gateway integration for marketplace bookings.

6.3 Legal and Compliance

- Data handling should align with applicable Ethiopian data protection norms and any relevant institutional (ASTU EDC) guidelines under which the project is incubated.

Appendix A: Hackathon Build Prioritization

Given a typical 24–48 hour hackathon window, the recommended build scope is a single, polished demo path rather than broad but shallow coverage:

Build (core demo path)

1. AI Video Mock Interview Simulator — one role, a fixed question set, near-real-time feedback on clarity, confidence, and STAR structure.
2. Skill Passport (basic) — an auto-generated summary page showing scores/feedback from the mock interview.
3. User Onboarding (minimal) — a 2-screen flow capturing name and target role/industry.
4. Progress Dashboard (lightweight) — a single growth visual (e.g., before/after score or a radar chart), seeded with one or two sessions.

Cut or fake for the demo

- Peer Language Exchange Matching — show a mockup screen rather than building live matching.
- AI Communication Practice Engine — fold into interview feedback or skip.
- Soft-Skills Drills — cut; low wow-factor for build effort.
- Marketplace, Verification System, Admin CMS, Notifications — present only as roadmap slides.

This scope proves technical execution (AI integration), market differentiation (Ethiopian-context feedback), and a tangible business-model output (the Skill Passport) without over-building Phase 2 features.